1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).
   Select * from orders where cnum =
   (Select cnum from customers where cname = ‘cisneros’)

```
d5_96305_sanchit> Select * from orders where cnum = (select cnum from customers where cname = 'cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.171 sec)

d5_96305_sanchit>
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.
   Select o.onum,c.cname, o.amt, c.rating
   From customers c
   Join orders o on c.cnum = o.cnum
   Where o.amt > (select avg(amt) from orders);

```
d5_96305_sanchit> Select o.onum,c.cname, o.amt, c.rating
    -> From customers c
    -> Join orders o on c.cnum = o.cnum
    -> Where o.amt > (select avg(amt) from orders );
+------+---------+---------+--------+
| onum | cname   | amt     | rating |
+------+---------+---------+--------+
| 3005 | Liu     | 5160.45 |    200 |
| 3011 | Clemens | 9891.88 |    100 |
| 3008 | Clemens | 4723.00 |    100 |
+------+---------+---------+--------+
3 rows in set (0.050 sec)

d5_96305_sanchit>
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.
   Select snum, sum(amt)
   From orders
   Group by snum
   Having sum(amt) > (select amt from orders order by amt desc limit 1);

```
d5_96305_sanchit> Select snum, sum(amt)
    -> From orders
    -> Group by snum
    -> Having sum(amt) > ( select amt from orders order by amt desc limit 1);
+------+----------+
| snum | sum(amt) |
+------+----------+
| 1001 | 15382.07 |
+------+----------+
1 row in set (0.073 sec)

d5_96305_sanchit>
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

   Select cname, cnum
   From customers
   Where rating > any (select c.rating
   From customers c
   Join salespeople s on s.snum = c.snum where s.sname = 'serres')

```
d5_96305_sanchit> Select cname, cnum
    -> From customers
    -> Where rating >= any ( select c.rating
    ->                                                                  From customers c
    ->                 Join salespeople s on s.snum = c.snum where s.sname = 'serres');
+----------+------+
| cname    | cnum |
+----------+------+
| Giovanni | 2002 |
| Liu      | 2003 |
| Grass    | 2004 |
| Cisneros | 2008 |
+----------+------+
4 rows in set (0.008 sec)

d5_96305_sanchit>
```

7. Extract all the orders of Motika

```
d5_96305_sanchit> select * from orders where snum = (select snum from salespeople where sname = 'motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.007 sec)

d5_96305_sanchit>
```

8. Find all the orders of salespeople servicing customers in London.

```
d5_96305_sanchit> select * from orders where snum = any (select s.snum
    -> from customers c join salespeople s on c.snum = s.snum
    -> where c.city ='london');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.010 sec)

d5_96305_sanchit>
```

6) Write a query that selects all orders for amounts greater than any for the customers in London.

```
d5_96305_sanchit> select * from orders where amt > any(select o.amt
    -> from orders o join customers c on o.cnum = c.cnum
    -> where c.city ='london');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.008 sec)

d5_96305_sanchit>
```

11. Select customers who have a greater rating than any other customer in Rome.

```
select * from customers ' at line 1
d5_96305_sanchit> select * from customers where rating > any(select rating from customers where city = 'rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.008 sec)

d5_96305_sanchit>
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
d5_96305_sanchit> select * from orders where amt > any (select amt from orders where odate = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.128 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
d5_96305_sanchit> select * from orders
    -> where amt < any(select o.amt
    -> from orders o join customers c on o.cnum = c.cnum
    -> where c.city = 'san jose');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.009 sec)

d5_96305_sanchit>
```

14. Select those customers whose rating are higher than every customer in London.

```
d5_96305_sanchit> select * from customers where rating > (select max(rating) from customers where city ='london');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.006 sec)

d5_96305_sanchit>
```

5) Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
d5_96305_sanchit> select *
    -> from salespeople p
    -> where p.city <> all(   select c.city
    ->                        from salespeople s
    ->                        join customers c
    ->                        on s.snum = c.snum
    ->                        where s.snum = p.snum);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.013 sec)

d5_96305_sanchit>
```

9) Find names and numbers of all salesperson who have more than one customer.

```
d5_96305_sanchit> select s.sname , s.snum
    -> from salespeople s
    -> where s.snum = any (select s.snum from salespeople s join customers c on s.snum = c.snum group by snum having count(*)
 > 1);
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.160 sec)

d5_96305_sanchit>
```

10) 10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
d5_96305_sanchit> select s.sname , s.snum , count(*)
    -> from salespeople s join customers c on s.snum = c.snum
    -> where s.snum = any (select s.snum from salespeople s join customers c on s.snum = c.snum group by snum having count(*)
 > 1)
    -> group by s.sname,s.snum ;
+--------+------+----------+
| sname  | snum | count(*) |
+--------+------+----------+
| Peel   | 1001 |        2 |
| Serres | 1002 |        2 |
+--------+------+----------+
2 rows in set (0.153 sec)

d5_96305_sanchit>
```